IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-086

1. Topic (12 words max)

AI Online Speech Disorder Screening & Therapy Assistant

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	no
Year	no

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Many people live with speech disorders such as stuttering, dysarthria, apraxia of speech, and articulation difficulties. Access to a speech-language pathologist is often limited by distance, cost, or long waiting lists. Without early screening and ongoing practice, people miss timely interventions and lose confidence in everyday conversations. Meanwhile, research has produced useful building blocks (e.g. - stuttering-event datasets like SEP-28k and motor speech corpora such as TORGO but there is no widely available online tool that can interview a user, summarize severity across key dimensions (fluency, rate, articulation, prosody), and then coach practice at home with clear progress tracking and privacy by design.

Low resource languages are especially underserved, though modern self-supervised speech models (e.g. XLS-R) and Indic ASR efforts show strong cross lingual transfer, focusing exclusively on English. Ethical guidance from the WHO also now clarifies how to build safe, transparent, and privacy-respecting AI in health.

Our goal is to create an online assistant that,

1. Runs a gentle, guided speech interview
2. Estimates four-level severity (Mild - Very Severe)
3. Provides real-time coaching (e.g. pacing, gentle onset, optional DAF)
4. Supplies personalized home practice with objective progress charts while keeping audio on device by default and using optional federated learning with differential privacy for model improvements.

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How the problem is tackled, in four small stages (overview),

1. Capture - A friendly online interview with spontaneous answers, a short reading, and a few sound drills in the user's chosen language (English.
2. Analyze - Evidence-based detection of disfluencies (repetitions, prolongations, blocks), articulation timing via MFA on reading tasks, and robust speech rate measures.
3. Classify and Coach - Map results to Level 1–4 with a confidence bar and give real time cues (plus optional DAF) that users can try immediately.
4. Practice and Track - A short, rotating practice plan + clear trend charts, data stays on device unless the user operates in to privacy-preserving learning.

User personas (primary & secondary),

- Primary - People (teens to adults) with stuttering, dysarthria, apraxia or articulation difficulties seeking at home screening and practice.
- Secondary - SLPs, parents/caregivers, teachers who may review progress PDFs and tailor in person therapy.

Impact

1. Social - Earlier screening, more confidence in conversations, better engagement at school/work. (Daily practice guided by the app lowers barriers to improvement)
2. Technical - Integrates MFA alignment, multi-label disfluency detection, low resource speech models, and on device + federated learning patterns within one workflow.
3. Economic - Reduces travel and clinic overhead for routine follow ups, supports SLPs with objective trend charts to focus live sessions.
4. Ethical and Legal: Aligns with WHO AI in Health principles (safety, transparency, fairness, privacy). Clear consent, data minimization, and opt-in analytics.
5. Environmental - Fewer unnecessary clinic trips and paper forms, digital summaries.

Problem Statement

There is no practical, privacy respecting online assistant that can interview users in English, summarize four-level speech-disorder severity, and deliver in the moment coaching with home practice and trend tracking, leaving many people without early support or continuous guidance. (We explicitly present this as screening/assistance, not a medical diagnosis.)

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6. Brief description of Existing Research and Systems (300 words max)

Stuttering & disfluency - Public datasets such as **SEP-28k** provide over 28k clips labeled for blocks, prolongations, and repetitions, enabling reliable event detection and multi label models. Reviews summarize features and deep learning methods but note generalization challenges and the need for clinically meaningful outputs that users understand.

Dysarthria and severity - Research using **TORGO** and **UA Speech** shows strong internal performance for dysarthria detection/severity with CNN/RNN hybrids, yet cautionary work demonstrates that some systems may inadvertently learn **recording condition artifacts** rather than true pathology, so **cross corpus validation** and confound checks are essential.

Alignment and articulation - For reading tasks, the **Montreal Forced Aligner (MFA)** produces accurate word/phone boundaries and has been shown to outperform some end-to-end ASR aligners for alignment accuracy making it a dependable foundation for articulation scoring when paired with alignment error aware reporting.

Low-resource speech tech XLS-**R** - Indic ASR initiatives demonstrate robust cross lingual transfer and practical ASR for multiple Indian subcontinent languages, which the system leverages for English speech tasks and interview support.

Mobile therapy and DAF -Trials and apps indicate that smartphone-based speech practice and **Delayed Auditory Feedback** can support fluency improvements for some users, provided they are used ethically and with clear guidance.

Ethics and governance - The **WHO** (2021, 2024 LMM guidance) provides clear principles (human oversight, transparency, safety, equity, and privacy) that we embed from day one in consent flows and data handling.

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7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

Proposed System Overview

A web and mobile AI assistant conduct a guided online interview (spontaneous answers + short reading + phoneme drills), then analyzes the audio to estimate fluency (repetitions, prolongations, blocks), speech rate & rhythm, prosody, and articulation clarity. A multitask classifier maps results to a four-level severity scale with a confidence bar and explains the main factors in plain language. The assistant then provides real time coaching (e.g. "slow down a little", "gentle onset"), an optional DAF toggle, and a personalized weekly practice plan with progress charts on device by default.

How the solution addresses the gap

1. Early remote screening without a clinic visit (clear, four level result + lay explanations).
2. Actionable in-the-moment help (pacing bar, gentle onset drills, DAF) rather than static tips.
3. English speech is supported through pre-trained SSL and ASR models, ensuring broad usability.
4. Privacy by design on device inference, optional federated learning with differential privacy if users opt in aligns with modern speech FL baselines.

System Perspective, (Input - Process – Output)

Inputs - User's mic audio from (a) spontaneous Q&A, (b) short reading passage, (c) phoneme drills, language choice.

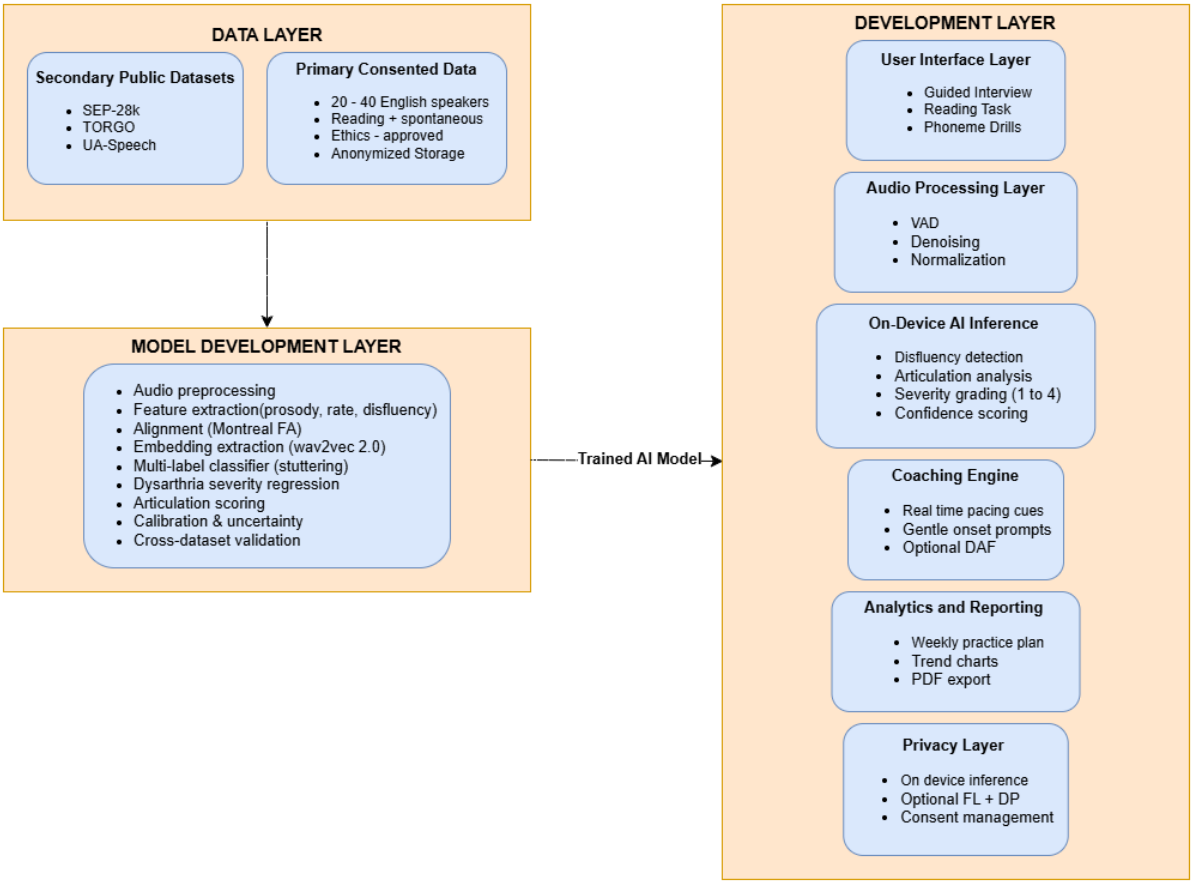
Process,

1. **Capture and Pre ASR** - VAD, denoise, level normalization, robust speech rate metrics.
2. **Alignment & Features MFA** - alignment (reading), extract features (prosody, pauses, timing, disfluency cues).
3. **Multi-task Severity** - Multi-label stuttering events (SEP-28k-style), dysarthria severity, articulation scoring with alignment-aware confidence, cross corpus checks to avoid confounds.
4. **Coach and Track** - Real-time prompts + DAF, weekly practice plan, trend charts, optional FL+DP.

Outputs - Severity Level 1 - 4 with confidence, brief explanation ("frequent sound repetitions + faster rate"), personalized exercises, trend report PDF for the user/SLP.

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Conceptual Diagram



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8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

Clinical alignment and ethics

We will map outputs to user-friendly language and standard clinical ideas (e.g. "disfluency events", "gentle onset"), with a screening/assistance disclaimer. Consent, transparency, and privacy first defaults follow WHO AI in health guidance (human oversight, safety, fairness, explainability, and privacy).

Corpora & benchmarks.

- Stuttering - SEP-28k (multi-label, blocks, prolongations, repetitions, interjections) for event detection.
- Motor speech - TORGO and UA Speech for dysarthria detection and severity, we will run cross corpus tests to reduce confounds from recording setups.
- Low resource speech - XLS-R pretrained SSL models to support English speech tasks.
- Alignment and rate - MFA for phone, word timing, robust speech rate metrics (error based) from recent prosody research.

Technical capacity.

- Feature and Modeling - wav2vec/XLS-R embeddings, multi-label sequence models for disfluency, regression/calibrated classifiers for severity, articulation scoring with alignment confidence.
- On device and privacy: TFLite /CoreML for mobile, optional Federated Learning + Differential Privacy following published baselines for end-to-end speech models.

Data collection & ethics.

We will collect small scale, consented English speech samples from 20 - 40 voluntary participants under controlled and semi realistic recording conditions. The recordings will include both reading and spontaneous speech to support UI validation, threshold tuning, model calibration, and usability testing. This primary data collection is intended solely for these purposes and **not** for large scale model training.

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The datasets we will use include public secondary datasets and primary data collected in house.

Public Secondary Datasets:

1. SEP-28k (https://www.kaggle.com/datasets/ikrbasak/sep-28k) for stuttering event detection
2. TORGO (https://www.kaggle.com/datasets/pranaykoppula/torgo-audio/data)for dysarthria severity modeling
3. UA-Speech (https://www.kaggle.com/datasets/aryashah2k/noise-reduced-uaspeech-dysarthria-dataset) for cross-dataset validation

These datasets are used exclusively for feature learning, multi-label model training, severity regression modeling, bias detection, and cross-corpus validation.

Feasibility.

Public datasets + modern SSL models + proven alignment tools (MFA) keep this project technically feasible within a final year scope, while leaving room for a small user pilot to test usability and clarity of feedback.

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9. Objectives and Novelty

Main Objective

Design, implement, and evaluate a privacy-preserving, English-language AI assistant that performs online speech interviews, classifies disorder severity (Level 1 to 4) across stuttering/dysarthria/articulation, and delivers adaptive therapy with real time feedback and progress analytics.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Student ID	*Your component of the project*	*Tasks you perform to achieve your component and overall objectives.*	*Your genuine contribution / What you can't observe in other research projects that have tackled the same problem*
P.A.S.S. Kusumsiri (IT22132628)	Noise-Robust Online Speech Screening and Self-Supervised Feature Extraction	1. Design and implement an AI-guided online speech screening interview using structured elicitation tasks (reading passages, picture description, and spontaneous speech) for comprehensive English speech assessment. 2. Develop a multi-stage audio preprocessing pipeline incorporating adaptive Voice Activity Detection (VAD), spectral noise suppression, and real-time speech quality	Existing speech disorder screening systems predominantly rely on clean, laboratory-recorded speech and standard ASR pipelines that degrade significantly when applied to disordered speech in real-world acoustic conditions. This research introduces an ASR-uncertainty-aware, noise-robust feature extraction pipeline that explicitly models the unreliability of automatic transcription on pathological speech. By fine-

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		assessment to handle unconstrained acoustic environments. 3. Fine-tune self-supervised speech foundation models (HuBERT / wav2vec 2.0 / Whisper) on disordered English speech corpora (TORGO, SEP-28k, UASPEECH) to extract disorder-sensitive acoustic representations. 4. Extract and validate a comprehensive fluency feature set (WPM, articulation rate, pause frequency, disfluency event density) using ASR confidence-weighted, error-aware metrics that account for ASR degradation on pathological speech. 5. Evaluate feature reliability across acoustic conditions (varying SNR levels, microphone types, background noise categories) using cross-corpus experiments to establish real-world robustness benchmarks.	tuning pre-trained speech foundation models (HuBERT / Whisper) on disorder-specific English corpora and combining ASR confidence-weighted fluency metrics with an adaptive speech quality gating mechanism, the system achieves clinically reliable disorder-sensitive feature extraction under adverse real-world conditions. Additionally, the multi-task elicitation protocol (reading passage, picture description, and spontaneous speech) enables disorder-discriminative feature capture across multiple speech production modalities within a single online session — a design not present in existing automated screening tools.

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E.Y. Randiw (IT220257086)	Alignment & Articulation Analytics	1. Implement phoneme-level forced alignment for detailed speech analysis. 2. Compute phoneme timing, duration variation, and articulation accuracy. 3. Detect misarticulations and timing irregularities linked to speech disorders. 4. Develop articulation sub-scores based on phonetic precision and consistency. 5. Incorporate alignment confidence scoring using boundary uncertainty.	Traditional articulation assessment relies heavily on manual clinical evaluation. Although forced alignment tools exist, they are rarely integrated into intelligent therapy systems with uncertainty modeling. This research introduces alignment-aware articulation scoring with interpretable sub-scores and boundary confidence measures. The approach bridges research-grade phonetic alignment with a clinic-friendly AI application.
I.L.S. Tharushan (IT21315114)	Multi-Disorder Severity Classification & Risk Prediction	1. Develop multi-label models to detect stuttering events such as repetitions, prolongations, and blocks. 2. Train severity estimation models for dysarthria and other speech disorders. 3. Perform cross-dataset validation and recording-condition bias analysis. 4. Implement calibrated probability outputs with uncertainty estimation.	Existing research typically addresses single speech disorders using binary classification methods. This research proposes a unified multi-disorder severity engine capable of detecting and grading multiple speech conditions simultaneously. By integrating bias-aware evaluation and uncertainty-calibrated predictions, the system enhances reliability and clinical safety in online screening environments.

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		5. Design a unified severity scoring mechanism across disorders.	
M.D.I. Perera (IT22056702)	AI Therapy Assistant & Privacy-Preserving Adaptive Learning	1. Develop a real-time AI therapy coach providing pacing and gentle onset feedback. 2. Implement delayed auditory feedback (DAF) as an optional therapy feature. 3. Generate personalized therapy dashboards and adaptive weekly plans. 4. Integrate federated learning for decentralized model improvement. 5. Apply differential privacy techniques to protect sensitive speech data.	Many existing systems focus solely on speech disorder detection without integrating therapeutic support. This research extends beyond screening by incorporating an AI-driven therapy assistant that provides real-time personalized feedback. Additionally, the integration of federated learning and differential privacy ensures secure and ethical handling of sensitive voice data. This creates a comprehensive, privacy-aware AI ecosystem for speech health.

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10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes		No	

b) The proposed topic exhibits novelty.

Yes		No	

c) The student group can successfully execute the proposed project.

Yes		No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes		No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

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